

● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

07

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $9x - 140^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A. $-18x + 280^\circ$
- B. $-18x + 640^\circ$
- C. $18x - 280^\circ$
- D. 180°

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is 5 square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

- A. 7
- B. 10
- C. 20
- D. 25

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has its center at $-5, 2$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

In triangle JKL , $\cos K = \frac{24}{51}$ and angle J is a right angle. What is the value of $\cos L$?

STUDENT-PRODUCED RESPONSE

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.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(4, 5)$, $(4, 7)$, and $(6, 5)$, respectively. Triangle LMN has vertices L , M , and N at $(4, 5)$, $(4, 7 + k)$, and $(6 + k, 5)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

- A. $90 - t - k^\circ$
- B. $90 - t + k^\circ$
- C. $90 - t^\circ$
- D. $90 + k^\circ$

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

Circle A has equation $x - 7^2 + y + 3^2 = 1$. In the xy -plane, circle B is obtained by translating circle A to the right 4 units. Which equation represents circle B?

A. $x - 7^2 + y + 7^2 = 1$

B. $x - 3^2 + y + 3^2 = 1$

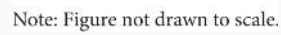
C. $x - 11^2 + y + 3^2 = 1$

D. $x - 7^2 + y - 1^2 = 1$

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$

GRID-IN GEOMETRY AND TRIGONOMETRY



STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Rectangles ABCD and EFGH are similar. The length of each side of EFGH is 6 times the length of the corresponding side of ABCD. The area of ABCD is 54 square units. What is the area, in square units, of EFGH?

- A. 9
- B. 36
- C. 324
- D. 1,944